

Electron Budget

Where the electrons in a Lewis structure come from, why the total is settled before you draw anything, and what to do when an atom is still short.

Name _____ Date _____ Period _____
 Demonstration: andresdbp.github.io/resources/demos/electron-budget/

1 Before you open anything

Answer from what you already know. Being right is not the point — the point is having something on paper that the demonstration can settle.

PREDICTION

Ammonia, NH_3 , and the ammonium ion, NH_4^+ . Which of the two has *more* valence electrons to place in its Lewis structure, and by how many?

Write one sentence saying how you got that.

2 Where the number comes from, and where it goes

Open the demonstration. The **Molecule** menu loads a skeleton; the **Where the budget comes from** panel shows the arithmetic; the **Atom by atom** panel reports each atom as you draw.

- From the **Molecule** menu choose **H_2O — Water**.
- Copy the arithmetic out of the **Where the budget comes from** table into the table below — its columns are **Source**, **each** and **e^-** .

Source	each	e^-
$1 \times \text{O}$		
$2 \times \text{H}$		
Budget		

- Now draw it. Click each dashed line once to make the two O–H single bonds.

- Click the oxygen once to add a lone pair, then once more for a second lone pair.
- Fill in a row of the table below after each of those stages, reading **Placed** and **Left** from the readouts, and the oxygen's **shell e⁻**, **formal charge** and **status** from **Atom by atom**.
- Press **Check** and write down what it tells you.

After this stage	Placed (e ⁻)	Left (e ⁻)	O shell e ⁻	O formal charge	O status
Nothing drawn yet					
Both O–H bonds made					
One lone pair on O					
Two lone pairs on O					

a. One column stays constant no matter what you draw, and one column reaches zero exactly when you are finished. Name them.

b. Click the oxygen a third time. What does the message under the drawing say, and what does it tell you to do instead?

3 The rule behind the number

Two separate things set the budget: which atoms are present, and the charge. Vary them one at a time.

- Load each molecule below from the **Molecule** menu and read **Budget**.
- In the last column write out the valence-electron sum yourself, the way the **Where the budget comes from** table does it.

Molecule	Budget (e ⁻)	Your arithmetic
CH ₂ O – Formaldehyde		
N ₂ – Nitrogen		
NH ₃ – Ammonia		
NH ₄ ⁺ – Ammonium ion		
PCl ₅ – Phosphorus pentachloride		

Now hold the atoms fixed and change only the charge. Type each of the following into the **Or type any formula** box, exactly as written, and read the **Budget** row at the foot of the **Where the budget comes from** table. The line just under that table restates it — for water it reads *for $H_2O - 8 e^- = 4$ pairs*.

You type	Budget (e^-)	What the line under the table says after the “=”
SO4+		
SO4		
SO4-		
SO4 ²⁻		
SO4 ³⁻		

Read the right number. The three large readouts — Budget, Placed, Left — always describe the molecule currently drawn on the board. Typing a formula the demonstration has no skeleton for updates the arithmetic table and the line beneath it, and leaves the board alone. Of the five species above only SO_4^{2-} is one you will meet; the other four are there to isolate what the charge alone does to the count.

a. Write the rule for the budget in your own words, in one sentence, covering both the atoms and the charge.

b. Two of the five rows in the first table have the same budget. Which two, and why does adding a hydrogen atom *and* a positive charge leave the total unchanged?

c. Go back to your prediction in part 1. Was it right? If not, say exactly what you assumed that turned out not to hold.

d. Two of your five typed rows report a number of pairs *plus one left over*. What has to be true of the budget for that to happen, and what does it force you to draw?

4 When “give everyone an octet” runs out

The usual shortcut is to hand every atom eight electrons and hope the total works out. Here is a molecule where it does not, and the repair that a test question expects.

1. Choose **CO₂ – Carbon dioxide**. Record the budget.
2. Click each dashed line once, giving two single bonds.
3. Click each oxygen three times, giving each one three lone pairs. Fill in the first row below.
4. Repair it: right-click (or press and hold) one oxygen once to take a lone pair back, then click the bond to that oxygen to make it a double bond. Do the same on the other side. Fill in the second row.
5. Press **Check**.

Stage	Placed (e ⁻)	Left (e ⁻)	C shell e ⁻	C status	C formal charge
Single bonds, 3 lone pairs per O					
After both repairs					

a. At the first stage the budget was fully spent and carbon was still short. Adding two more pairs would have fixed carbon. Why is that not allowed?

b. Compare **Placed** before and after the repair. Where did the electrons in the second C–O bond come from?

Now a case where the octet is reachable and still rejected.

1. Choose **BF₃ – Boron trifluoride**, press **Show accepted answer**, and tick **Show formal charges on the drawing**. Record boron's **shell e⁻** and **status**.
2. Take one lone pair back from one fluorine and spend it as a second B–F bond. Press **Check**.

c. Boron in the accepted answer: shell e^- = _____, status = _____. After you made the double bond, what formal charges does the message report, and on which atoms?

d. Both drawings use all 24 electrons. Give the reason the electron-poor boron is the accepted structure anyway.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it, check yourself, and if a number is off, write down where the count went wrong.

1. Find the electron budget for the chlorate ion, ClO_3^- . State how many pairs that is.

2. The phosphate ion, PO_4^{3-} , is drawn with phosphorus in the centre, four single P–O bonds and three lone pairs on every oxygen. Find the budget, show that this drawing spends it exactly, and give the formal charge on phosphorus and on each oxygen. Check that they add to the ion's charge.

3. Hydrogen cyanide is HCN, with carbon in the middle. Find the budget, then give a structure in which carbon and nitrogen both reach eight and hydrogen reaches two. State how many of the electrons end up in bonds and how many end up as lone pairs.

4. In the demonstration, phosphorus in PCl_5 is labelled *expanded, 10* and is accepted. A drawing that put 10 electrons on nitrogen would be labelled *impossible*. Explain the difference between the two atoms, and state the counting rule that never bends.

6 On your own

Nothing here is graded. The demonstration holds more than this worksheet uses.

- Load **NO — Nitric oxide**. Its budget is odd, and **Place single electrons** switches itself on. Work out for yourself why no arrangement of pairs can ever total 11, and why a triple bond does not rescue it.
- Build the accepted structure for **O₂ — Oxygen** and read the message. It satisfies every rule in this unit and still disagrees with a laboratory magnet. Decide what that costs the model.
- Load **NO₃⁻ — Nitrate ion** and build it three times, putting the double bond on a different oxygen each time. Turn on **Show formal charges on the drawing** and compare. Which drawing is the real ion?
- Type a formula containing a transition metal, such as **FeCl₃**. Read what the demonstration refuses to do, and work out why counting valence electrons this way stops working outside the main group.